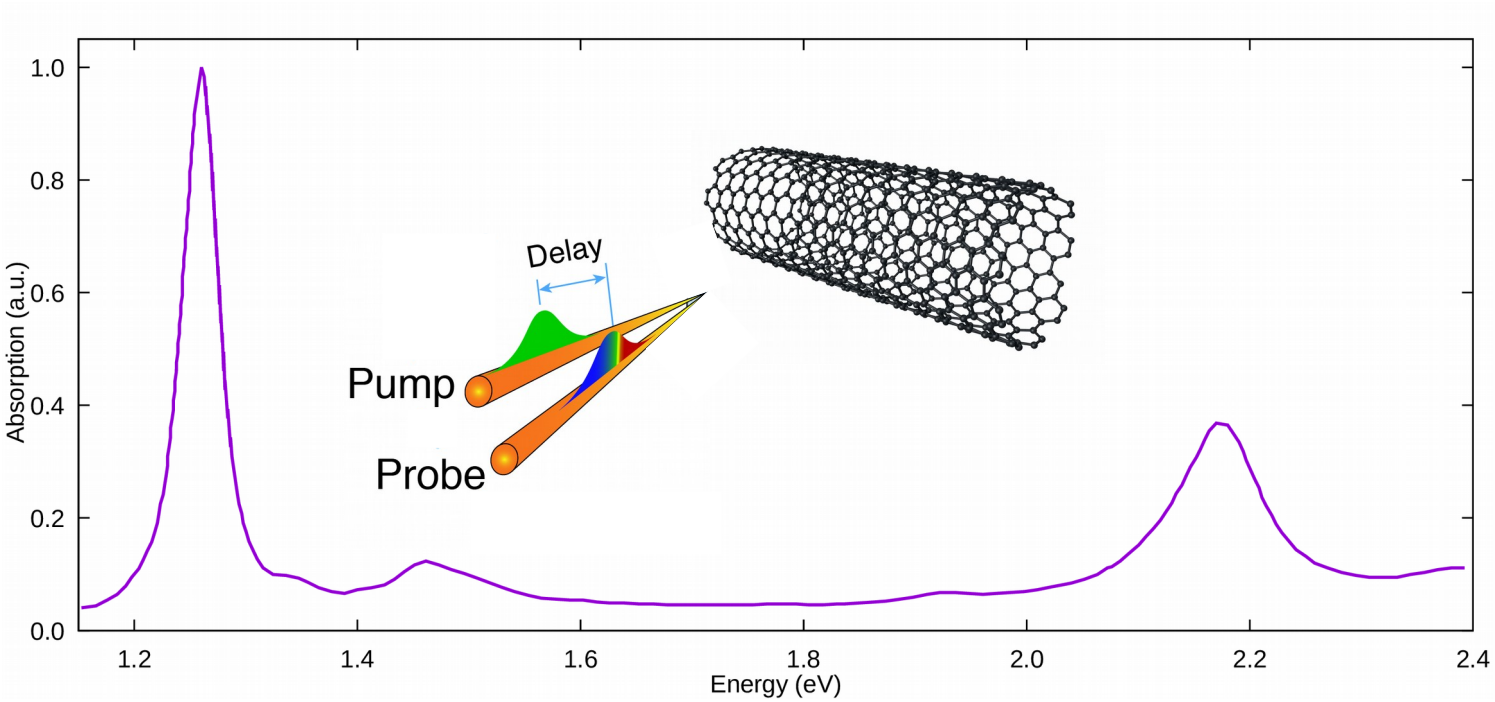
Exciton- and phonon-mediated electron ultrafast dynamics in semiconducting carbon nanotubes

Dr. Stefano Dal Forno



Motivation

Carbon nanotubes (CNTs) are quasi-1D systems that can be semiconducting or metallic. They are promising candidates for a number of technological applications:

- opto-electronic devices (field effect transistors, high-speed signal carriers, …);
- chemical or gas sensors (functionalised CNTs, …);
- strong and flexible conducting devices (transparent conductive films, flexible displays, …);
- compound materials for the improvement of thermal, mechanical and electronic properties.

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For <u>nanoelectronic applications</u>, a microscopic understanding of the electron, hole, phonon and exciton dynamics in CNTs is needed.

The quantum Boltzmann equation

To study the dynamics and the interactions between charge carriers, phonons and excitons in CNTs, we invoke the quantum Boltzmann equation:

$$\frac{\partial f_n}{\partial t} + \mathbf{v}_n \cdot \frac{\partial f_n}{\partial \mathbf{r}} - \frac{e}{\hbar}\left[\mathbf{E} + \frac{\mathbf{v}_n}{c} \times \mathbf{B}\right] \cdot \frac{\partial f_n}{\partial \mathbf{k}} = \left(\frac{\partial f_n}{\partial t}\right)_{coll}$$

$$\frac{\partial f_n}{\partial t} = \frac{\partial f_n}{\partial t}(t, \mathbf{r}, \mathbf{k}) \qquad \mathbf{v}_n(\mathbf{k}) = \frac{1}{\hbar}\frac{\partial \epsilon_n(\mathbf{k})}{\partial \mathbf{k}}$$

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A commonly used approximation is the quantum Fokker-Planck equation:

$$\left(\frac{\partial f_1}{\partial t}\right)_{12-34} = \sum_{\mathbf{G}} \int\int\int_{V_{BZ}^3} d\mathbf{k}_2 d\mathbf{k}_3 d\mathbf{k}_4 \left|g_{34}^{12}\right|^2 \delta_{\mathbf{k}}\, \delta_\epsilon\, P_{34}^{12}$$

$$\delta_{\mathbf{k}} = \delta(\mathbf{k}_1 + \mathbf{k}_2 - \mathbf{k}_3 - \mathbf{k}_4 + \mathbf{G})$$

$$\delta_\epsilon = \delta(\epsilon_1(\mathbf{k}_1) + \epsilon_2(\mathbf{k}_2) - \epsilon_3(\mathbf{k}_3) - \epsilon_4(\mathbf{k}_4))$$

$$P_{34}^{12} = [(1-f_1)(1-f_2)f_3 f_4 - f_1 f_2 (1-f_3)(1-f_4)]$$

(electron-electron scattering example)

Solving the quantum Fokker-Planck equation

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Very difficult because:

- It is a 12 dimensional integral;
- It is a 4th order operator due to the population factor;
- Energy and momentum deltas are difficult to handle;
- Very difficult to converge.

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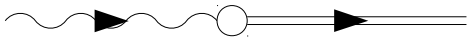
Our approach:

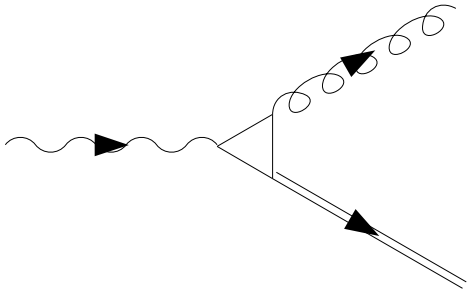
- Project the integrand onto a set of localised basis functions with compact support:
  → The scattering integral become a sparse tensor.
- Basis functions are second order polynomial:
  → Fewer k point needed. Exact inversion of Dirac deltas.
- Analytic inversion of Dirac deltas:
  → The dimensionality of the integral is reduced. Exact energy and momentum conservation.

Practical example: absorption spectrum of (6,5) CNTs

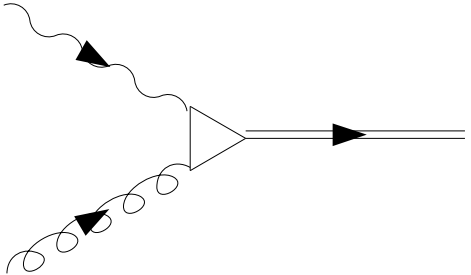
We consider the semiconducting (6,5) CNT and compute the collision integrals for three different scattering processes:



Direct photon absorption into an exciton



Photon absorption into an exciton, via phonon emission



Photon absorption into an exciton, via phonon absorption

Photon

Phonon

Exciton

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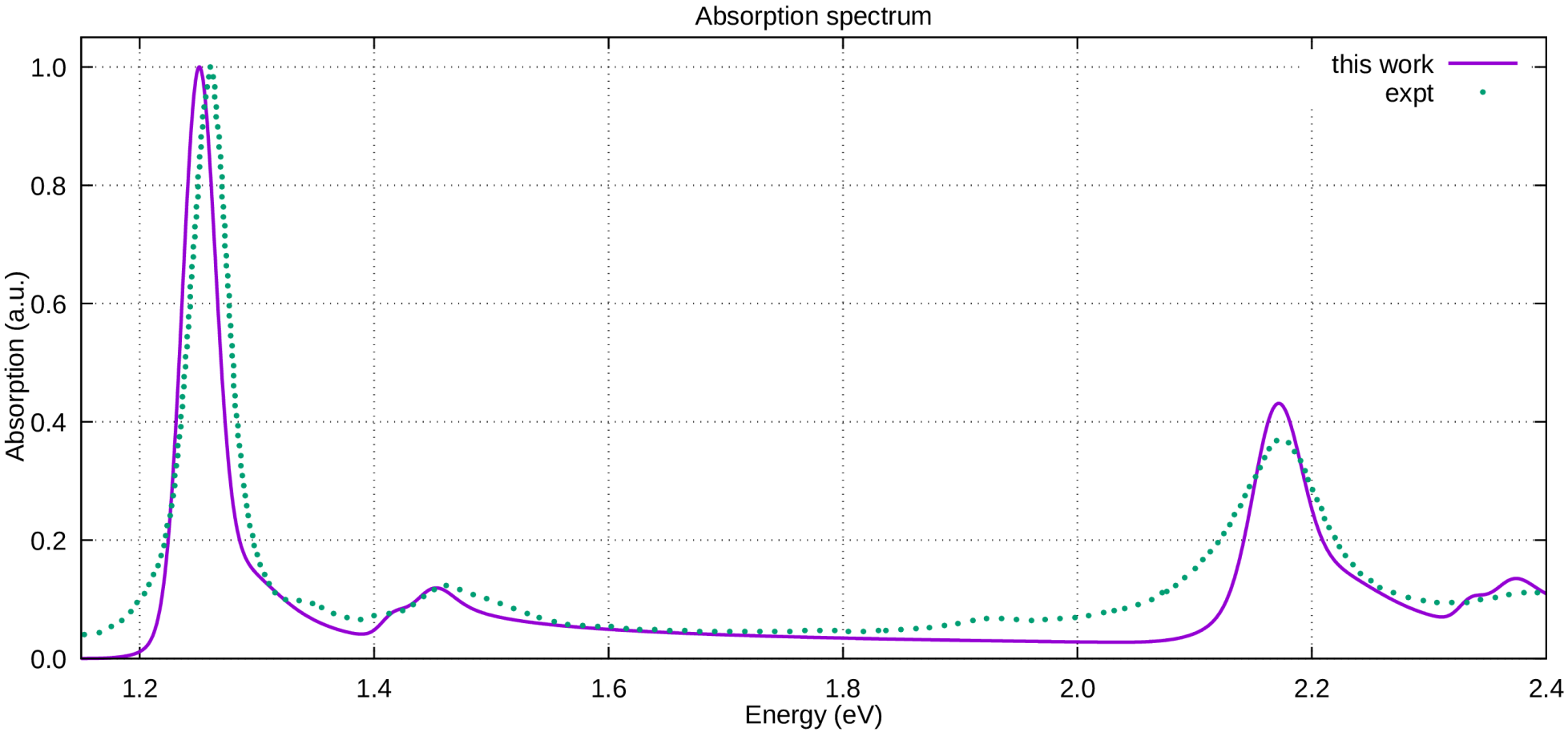

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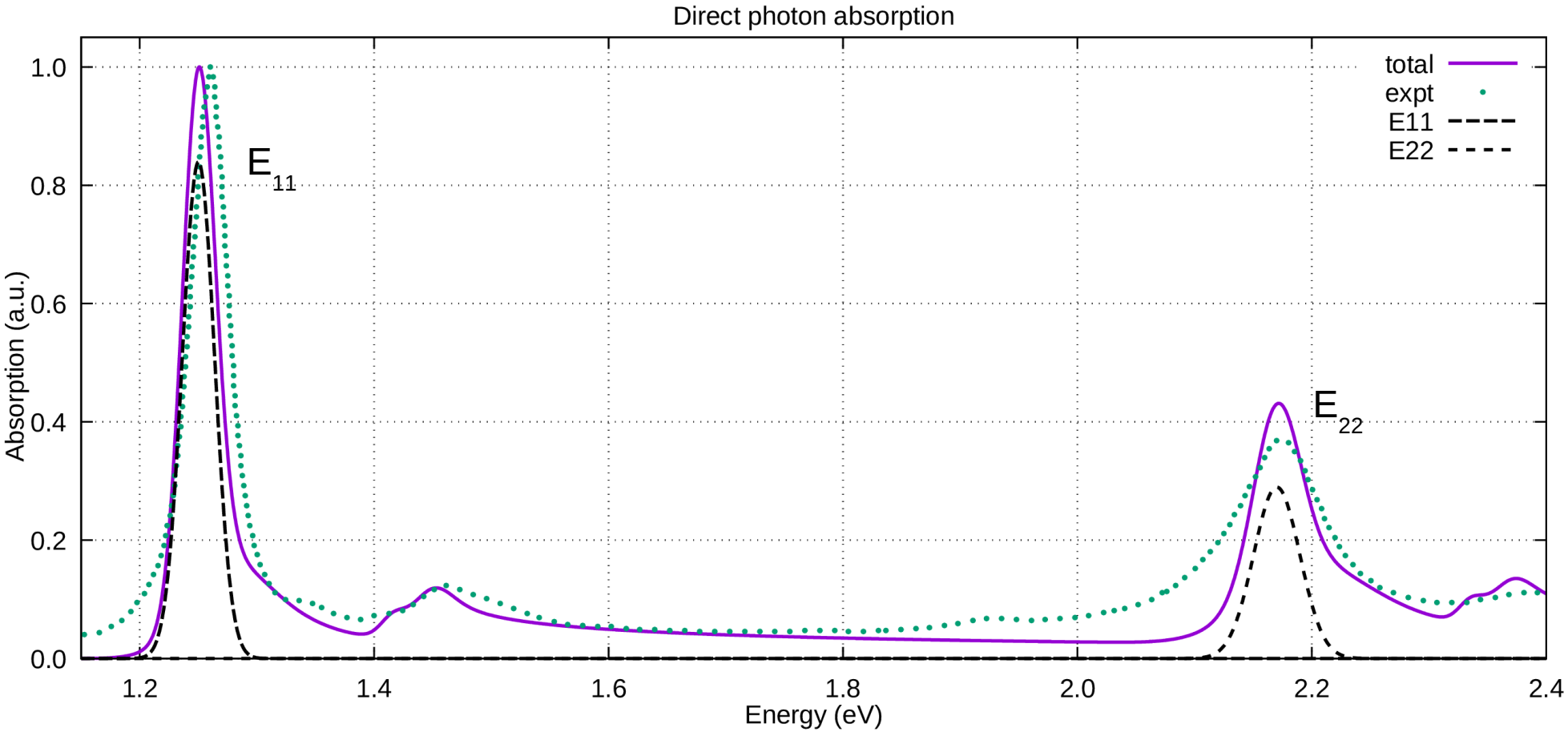

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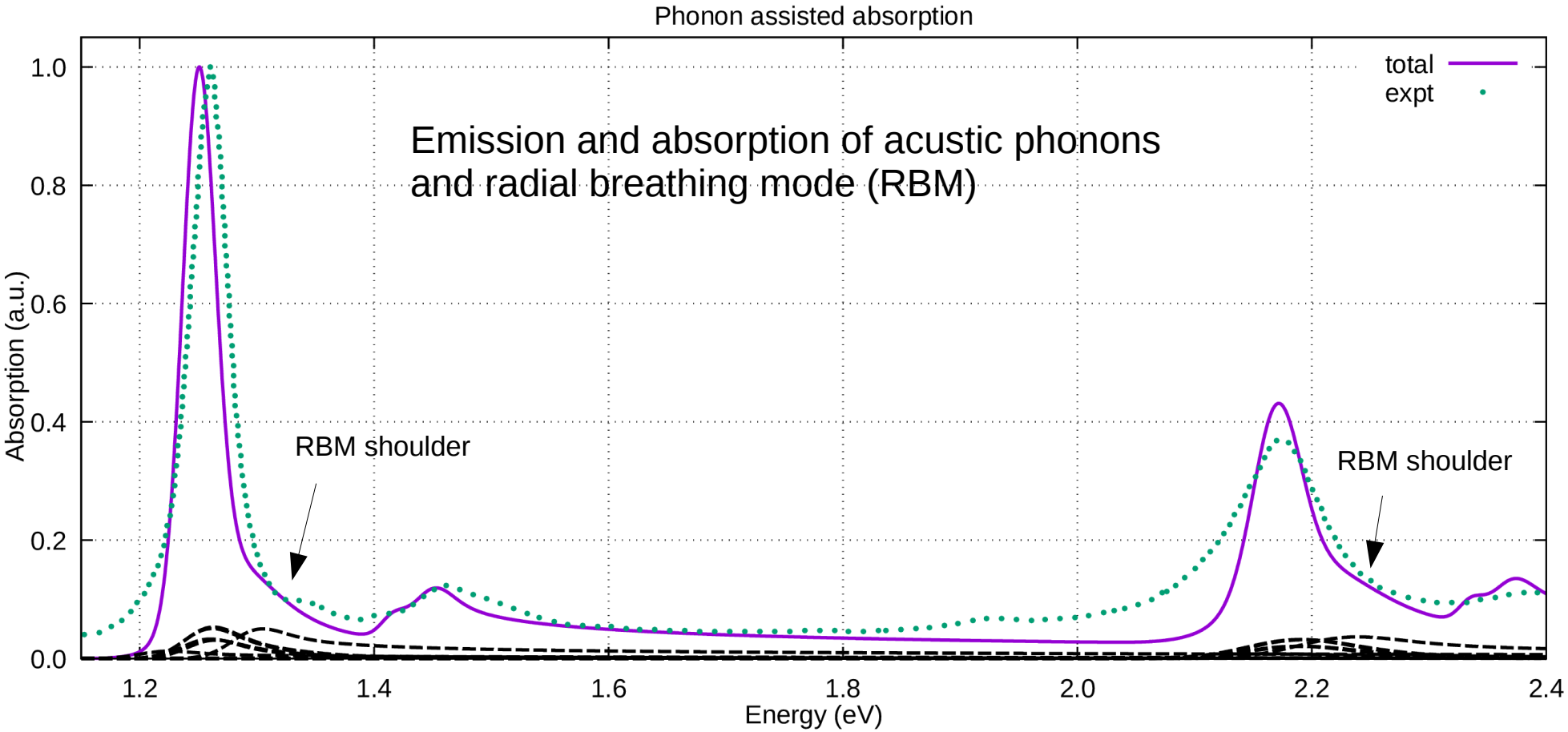

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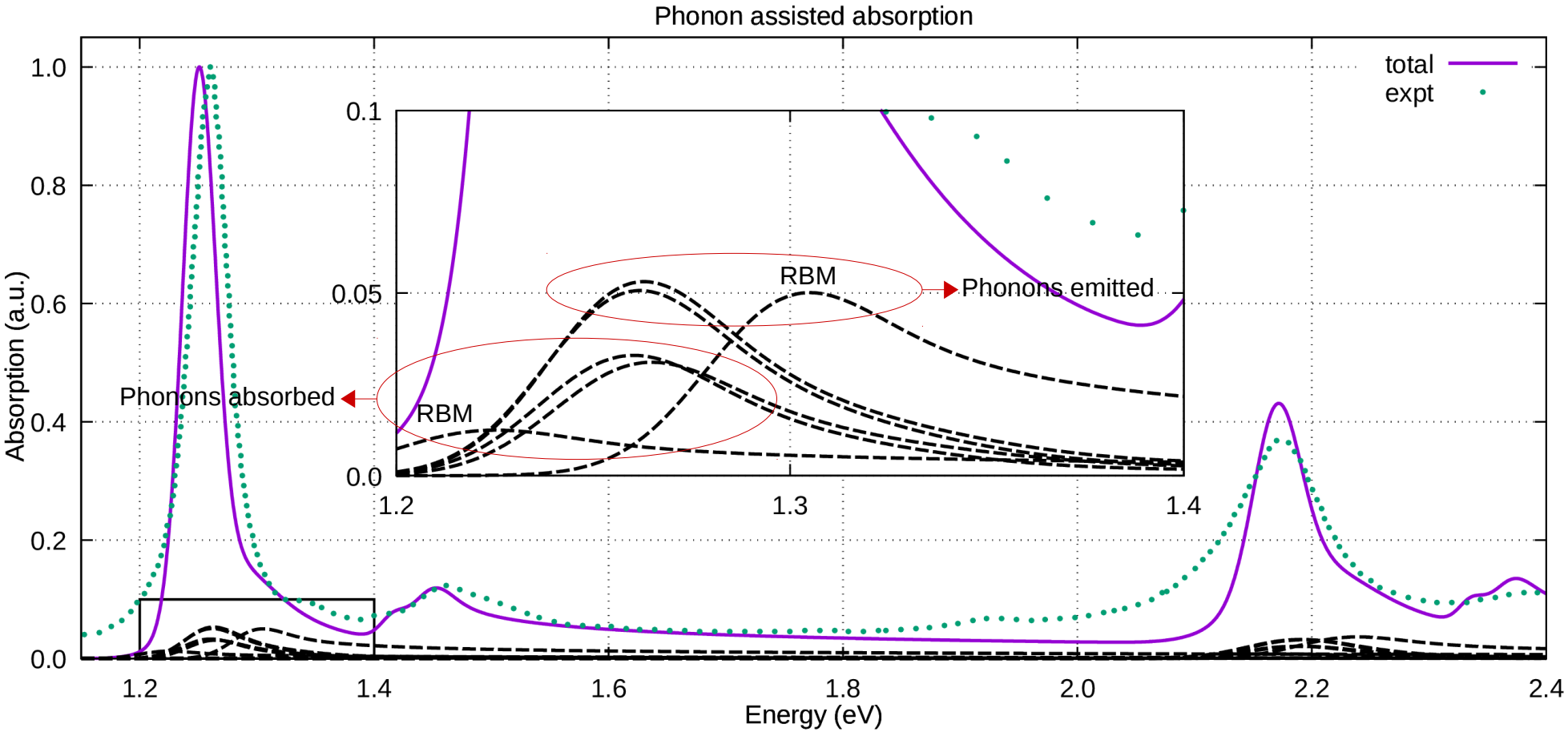

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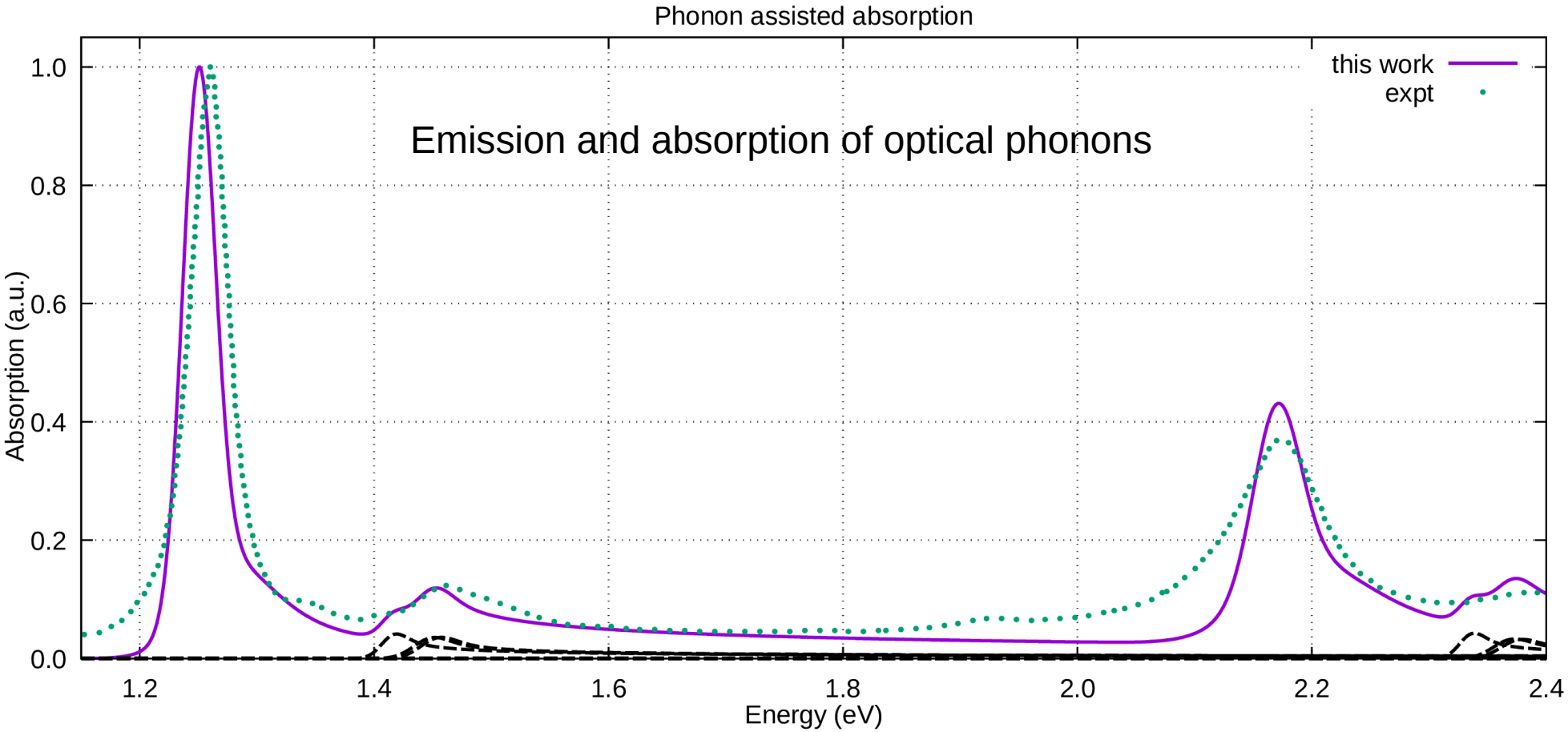

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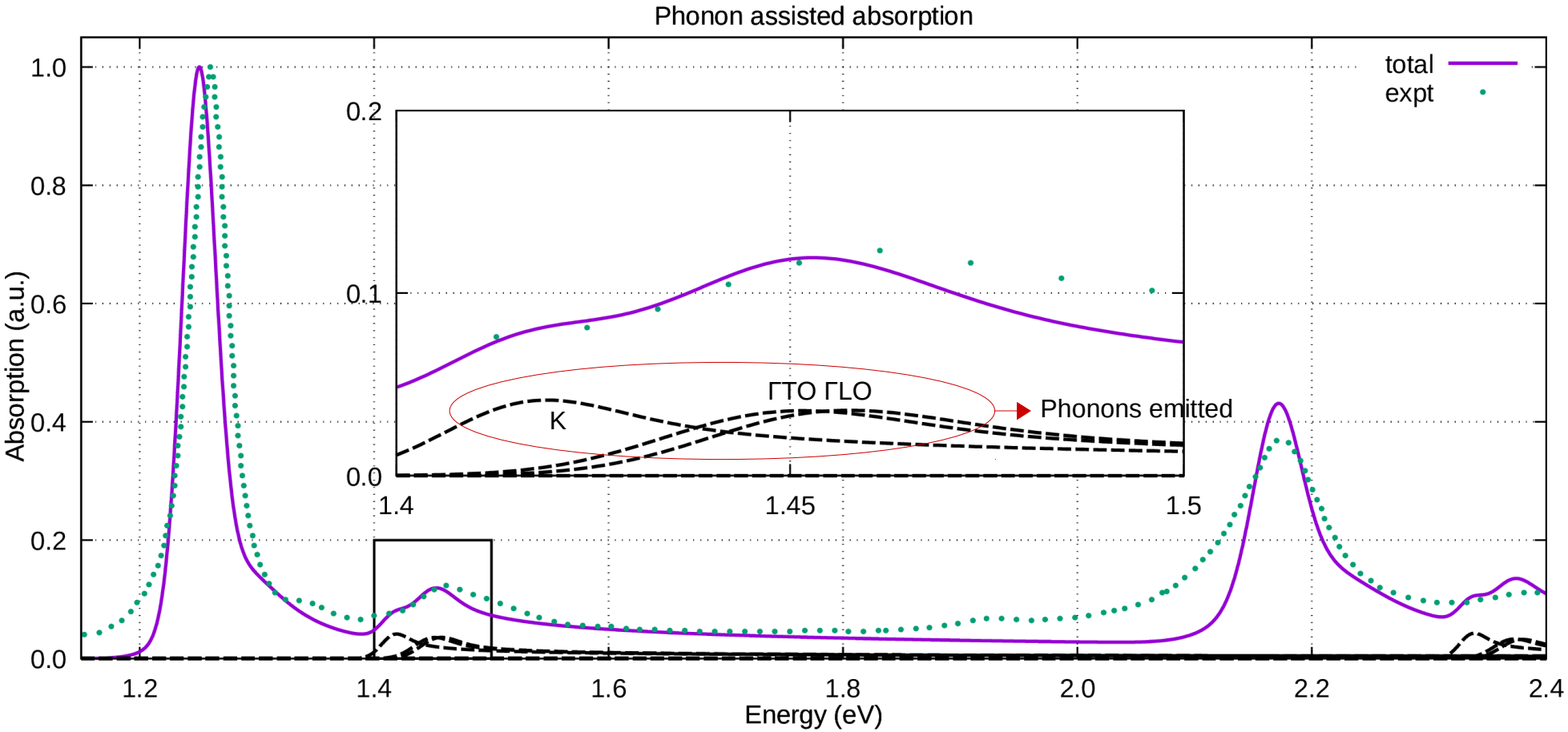

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Conclusion

- Exact solution of the Boltzmann scattering integrals;
- Capability of simulate true out of equilibrium systems;
- Excellent agreement with experiments.

- The matrix elements must be provided from third party source;
- The quantum Fokker-Planck equation breaks down for strongly interacting quasiparticle (e.g. electron-electron interaction).

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Conclusion